Lab Solution: Delta Lake

Creating a Delta Table

Run the cell below to create the **persons** Delta Table, and apply some operations on it.

4

```
USE CATALOG hive_metastore;

CREATE OR REPLACE TABLE persons
  (id INT, name STRING, age INT);

INSERT INTO persons
VALUES
  (1, "Tom", 18),
  (2, "Kumar", 25);

INSERT INTO persons
VALUES
  (3, "Ali", 50),
  (4, "Sandra", 35);

INSERT INTO persons
VALUES
  (5, "Eric", 28),
  (6, "Salma", 42);

UPDATE persons
SET age = age + 10
WHERE id = 1;

DELETE FROM persons
WHERE name = "Eric";
```

Q1 - Querying Delta Lake table

Query the data in the **persons** table using **SELECT** statement

6

```
-- Answer
SELECT * FROM persons
```

Q2 - Checking table history

Review the history of the table transactions using the **DESCRIBE HISTORY** command

8

```
-- Answer
DESCRIBE HISTORY persons
```

Q3- Checking table metadata

Review the basic metadata information of the table using the **DESCRIBE DETAIL** command

```
10
--Answer
DESCRIBE DETAIL persons
```

Q4- Exploring table directory

Explore the table directory using the **%fs** magic command.

Hint: get the table location from the above metadata information

```
12
%fs ls 'dbfs:/user/hive/warehouse/persons'
```

Q5- Exploring the transactions log

Explore the **_delta_log** subfolder in the table directory

```
14
%fs ls 'dbfs:/user/hive/warehouse/persons/_delta_log'
```

```
15
```